Making Tail Constraints Bind: Two Silent Failure Modes in CVaR-Constrained Portfolio RL

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Abstract

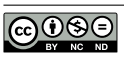
A portfolio allocator trained by reinforcement learning (RL) to maximise return has no reason to avoid large losses. The usual safeguard caps Conditional Value-at-Risk (CVaR), the average loss in the worst weeks, through a Lagrangian penalty. We show that this safeguard can fail silently in two ways. First, when the penalty and the reward sit on different scales, through an unstandardised cost advantage or a dual step mis-scaled to the CVaR excess, the penalty weight stays at zero and the budget is never enforced, even as tail metrics keep improving. Second, the evaluation protocol can decide the winner. On a broad cross-asset universe, a single stress window ranks the learned allocator above every classical optimiser, a rolling walk-forward reverses that ranking, and the reduction in tail risk holds under both. Plotting the multiplier during training exposes the first failure, and rolling out-of-sample folds expose the second. Fixing the scale makes the penalty bind in training, but the budget is still breached in most out-of-sample weeks. We therefore add a decision-time filter. Before each trade, it forecasts the portfolio's CVaR from trailing data and, when the forecast exceeds the budget, moves the portfolio toward minimum variance. At matched settings, the filter cuts the breach rate from 75.1% to 7.4%, close to the classical optimisers, at a Sharpe ratio comparable to rolling minimum variance. Across a rolling walk-forward on two ETF universes with weekly data and trading costs, the reduction in tail risk is statistically significant, including when folds are clustered by calendar and after Benjamini–Hochberg control. What the constraint delivers is protection against deep losses, and that protection survives multiple-testing correction.

CCS Concepts

• **Computing methodologies** → **Reinforcement learning**; • **Applied computing** → *Economics*.

Keywords

portfolio allocation, reinforcement learning, Conditional Value-at-Risk, drawdown control, constrained MDP, differentiable optimisation



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1 Introduction

A portfolio allocator decides, each week, how to split capital across assets. Reinforcement learning (RL) can learn such a rule from data, but a rule trained only to earn return has no reason to limit its worst losses. In the week of 16 March 2020 the S&P 500 fell about 15% over five sessions and the VIX closed at a record 82.69. An allocator trained without a loss limit kept holding equities, whereas an otherwise identical allocator with one had already rotated into Treasuries and cash before the trough, and lost about half as much. Its drawdown was 7.7% in March 2020 against 16.3%, and 11.8% against 20.6% over the whole 2020–2024 window (Figure 1, Table 3).

The limit we use is Conditional Value-at-Risk (CVaR), the average loss over the worst few percent of weeks [24]. The standard way to impose such a limit in RL is a Lagrangian penalty [2], in which the learner pays a price λ for each unit of CVaR above its budget, and λ rises automatically while the budget is exceeded. We show two ways in which this machinery can appear to work while the limit is never enforced.

Failure 1. The penalty never switches on. If the penalty signal is on a much smaller scale than the reward signal, λ stays near zero and the budget exists only on paper. Safe-RL work already knows that λ responds slowly and depends on its step size [26], and libraries such as OmniSafe standardise the cost signal by default [17]. In portfolio allocation the failure is easy to miss, because the raw cost signal is $O(10^{-3})$ against a unit-scale reward, and because tail metrics improve anyway through other parts of the pipeline. The problem only shows up if one plots λ over training (Section 5).

Failure 2. The test protocol picks the winner. Finance has long known that optimised portfolios which look strong in one sample often fail out of sample [5, 11], but RL portfolio studies usually report a single test window. On a 31-asset universe, one stress window ranks our learned allocator above every classical optimiser, while a 20-fold rolling walk-forward reverses that ranking (Section 7.8). "Beat-the-optimiser" results can therefore be artefacts of the split.

The fix. Even a correctly scaled penalty only discourages breaches on average, so the budget is still breached out of sample. We therefore check every proposed portfolio before it trades,

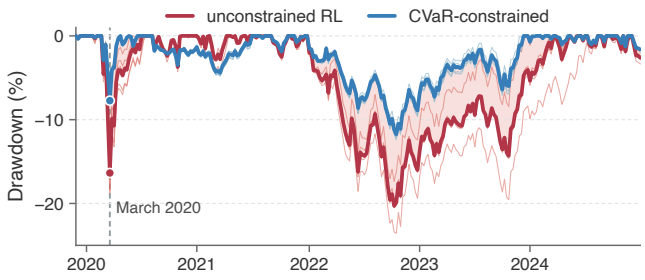


Figure 1: Drawdown on the 2020–2024 stress window. Thick lines follow the five-seed mean return path and thin lines the individual seeds.

in the manner of a safe-RL safety layer [10]. If its estimated forward CVaR exceeds the budget, the portfolio is shrunk toward a minimum-variance portfolio until it complies.

Contributions.

(1) Two named failure modes, with diagnostics and fixes. The first is a *Lagrangian constraint-coupling failure*, wherein a scale mismatch between the reward and constraint channels leaves λ inert, through an unstandardised cost advantage in a model-free actor–critic and through a dual step mis-scaled to the CVaR excess in a differentiable allocator (Section 5). The second is an *evaluation-protocol reversal* between a single stress window and a rolling walk-forward (Section 7.8).
(2) A differentiable CVaR-constrained allocator whose tail effect survives multiplicity control. A return objective plus a differentiable CVaR penalty is back-propagated through the rollout of a leak-free, tested environment (Section 4), and the decision-time filter cuts out-of-sample breach from 75.1% to 7.4% at matched dual settings. The tail-risk reduction holds on a 40-fold, two-universe walk-forward under Benjamini–Hochberg and (for Sharpe) Deflated-Sharpe control (Section 7.4), and a regime-switching hybrid reaches a walk-forward Sharpe of 1.316 vs 1.209 for pure minimum variance, with overlapping bootstrap intervals (Table 5).
(3) An evaluation protocol built to expose this kind of artefact. Deterministic optimisers and three tuned model-free RL families are compared under a two-universe rolling walk-forward with Wilcoxon, Benjamini–Hochberg and Deflated-Sharpe control, which is what surfaces the reversal in Contribution 1.

Our central claim is that the constraint protects against deep losses. We contribute a constraint that measurably binds and lowers tail losses, a decision-time filter that keeps a learned policy within its budget out of sample, and two failure modes that other studies of constrained RL in finance can check for.

2 Background and related work

Mean–variance optimisation and its inputs. Mean–variance optimisation (MVO) [22] is sensitive to input estimates. Decision-focused learning trains return predictors to improve downstream MVO decisions rather than raw forecast accuracy [20], and mean–variance-efficient collaborative filtering blends user preference with portfolio theory [9]. Both improve the inputs to a variance-based objective. We instead constrain a tail risk measure on a learned policy.

Conditional Value-at-Risk. CVaR_α is the expected portfolio loss on the worst $(1-\alpha)$ fraction of outcomes, the average of losses exceeding the α-quantile (VaR_α). Unlike variance, it ignores the upside, and unlike VaR, it averages across the whole tail. Rockafellar and Uryasev [24] give the tractable form

$$\mathrm{CVaR}_\alpha(L) \;=\; \min_{\zeta\in\mathbb{R}}\left\{\zeta + \frac{1}{1-\alpha}\,\mathbb{E}[\max(L-\zeta,\,0)]\right\}, \qquad (1)$$

where L denotes the portfolio loss and the inner expectation is over the empirical loss distribution. The minimum in (1) is attained at $\zeta = \mathrm{VaR}_\alpha$, and the representation admits both a convex linear program (LP), included as a baseline, and a differentiable penalty via `topk`.

RL for portfolio management. Most RL portfolio papers train on a single historical window, test on one held-out split, and compare against equal weight or buy-and-hold. EIIE [18] compares twelve classical strategies on three ≈50-day crypto back-tests, and SARL [29] reports higher return on a fixed period without cross-validation. FinRL [21] supports rolling retraining but demonstrates it on a single split with no significance tests. Huang [16] applies continuous-time RL to asset–liability management, although with a quadratic surplus objective rather than a tail constraint. None of these uses CVaR as a constraint, seeds multiple runs, or applies statistical testing. Empirical finance has long documented how the selection of backtests inflates reported performance [3, 15, 28] and that optimised portfolios rarely beat naive ones out of sample [11], and our protocol carries those controls over to RL portfolio allocation. We are not aware of prior work that subjects RL portfolio allocation to a 40-fold, two-universe walk-forward with Wilcoxon and Benjamini–Hochberg control, which is the protocol that exposes the reversal artefact.

Direct reinforcement and pathwise gradients. Moody and Saffell [23] trained a recurrent trading policy by back-propagating the *differential Sharpe ratio* directly through the return function. Buehler et al. [7] (Deep Hedging) extend this to neural derivative hedging, back-propagating CVaR as a risk-aversion *utility* through a simulated trajectory. Our allocator uses the same pathwise gradients, but replaces the risk-aversion coefficient with an explicit CVaR *budget constraint* enforced by a dual variable. Model-free policy gradients (Advantage Actor-Critic, A2C, and Proximal Policy Optimization, PPO [25]) and off-policy actor–critics (Soft Actor-Critic, SAC [14]) ignore this differentiable structure. Our allocator exploits it, which matters at this sample size (887 weekly observations, Section 7.7). ProtoHedge [13] similarly makes such policies auditable via a prototype-similarity policy.

CVaR-constrained RL. Chow et al. [8] derive the Lagrangian gradient for CVaR-constrained CMDPs. Tessler et al. [27] (RCPO) impose general constraints by a Lagrangian penalty and Ying et al. [30] (CPPO) constrain CVaR, both in robotics. Lagrange multipliers are known to respond slowly and to depend on their step size [26], and safe-RL libraries standardise the cost advantage by default [17]. None of this work documents the resulting *constraint-coupling failure* in portfolio allocation. There the raw

CVaR cost is $O(10^{-3})$ against a unit-scale reward advantage, a mismatch that robotics benchmarks would not expose. We pose allocation as a constrained MDP [1, 2] and show that a well-scaled dual is what makes the constraint bind. The closest concurrent work is Dong [12] (ClauseLens), a CVaR-constrained RL system for reinsurance pricing. Our work is complementary, since the failure modes we describe apply to any CVaR-constrained financial RL system.

3 Problem formulation

Markov Decision Process (MDP). Each week t, the agent observes a state s_t and chooses post-trade weights w_t on the long-only simplex. The state holds each asset's 26-week momentum and volatility, the current weights, the drawdown, a rolling 95% CVaR estimate and, optionally, exogenous macro and factor features. The environment projects w_t onto the admissible set $\mathcal{W}$ (max weight 0.40, turnover cap 0.50, optional cash floor), charges a proportional cost $c\|w_t - w_{t-1}\|_1$ with $c = 5$ bps, and realises the net return $r_t = w_t^\top \rho_t - c\|w_t - w_{t-1}\|_1$, where ρ_t are next-week asset returns. Features at t use only $\rho_{<t}$ (no look-ahead, enforced by tests).

Constrained objective (CMDP). A *constrained Markov Decision Process* (CMDP) [2] adds to the MDP one or more inequality constraints that the policy must satisfy on average. We maximise risk-adjusted return subject to a tail budget, $\max_\pi J(\pi)$ s.t. $\mathrm{CVaR}_\alpha(\text{loss}) \le d$ with $\alpha = 0.95$. The Lagrangian relaxation turns the constraint into a penalty, $\mathcal{L}(\pi, \lambda) = J(\pi) - \lambda\,(\mathrm{CVaR}_\alpha - d)$. The dual variable $\lambda \ge 0$ (the *dual*, or multiplier) is raised whenever the constraint is violated and lowered when slack remains. In equilibrium, λ is the shadow price of tail risk, so a higher λ means that the policy has been pushing against its tail budget. A portfolio is *over budget* when its $\mathrm{CVaR}_{0.95}$ (the mean weekly loss in its worst 5% of weeks) exceeds d (1.2% unless stated). It is measured on the sampled window in training, on the trailing 52 realised weeks at test time, and on a trailing forecast in the filter. The *breach rate* is the share of test weeks over budget (5% tolerated). We mostly report $\mathrm{CVaR}_{0.99}$, a deeper tail than the one constrained and so a stricter test. CVaR and drawdowns are fractions of wealth.

4 Method

Differentiable allocator. The actor is an MLP mapping s_t to simplex weights via softmax. The environment's reward and a sample CVaR are both differentiable in w, so over randomised training windows we back-propagate the objective

$$\mathcal{L} = -\,\mathrm{Sharpe}(\text{net returns}) \;+\; \lambda\,\mathrm{ReLU}(\widehat{\mathrm{CVaR}} - d) \;+\; \kappa\,\text{turnover}. \tag{2}$$

Here $\mathrm{Sharpe} = \bar{r}/\sigma_r$ is the window's mean net return divided by its standard deviation (reward per unit of risk), $\widehat{\mathrm{CVaR}}$ is the mean tail loss over the worst $(1-\alpha)$ fraction of window net returns (a differentiable tail mean via `topk`), and d is the pre-specified tail budget. This *pathwise* gradient, taken through the portfolio return itself, needs far fewer samples than *score-function* RL (PPO, A2C), which estimates it from sampled actions (Section 7.7).

Pathwise gradient and where the constraint acts. Because every term in $\mathcal{L}$ (2) is a differentiable function of the rollout, the policy is trained by exact back-propagation through the simplex map and the reward, without a score-function estimator. With net returns $r_t = w_t^\top \rho_t - c\|w_t - w_{t-1}\|_1$ and tail set $\mathcal{T} = \{t \mid r_t \text{ among the worst } \lceil (1-\alpha)T \rceil \text{ of the window}\}$, the gradient is

$$\frac{\partial \mathcal{L}}{\partial \theta} = \sum_t \Big(\underbrace{-\frac{\partial\,\mathrm{Sharpe}}{\partial r_t}}_{\text{all weeks}} \;-\; \underbrace{\tfrac{\lambda}{|\mathcal{T}|}\,\mathbf{1}[\widehat{\mathrm{CVaR}} > d]\,\mathbf{1}[t \in \mathcal{T}]}_{\text{tail weeks only}} \Big)\, \frac{\partial r_t}{\partial w_t}\,\frac{\partial w_t}{\partial \theta}, \tag{3}$$

where $\partial w_t / \partial \theta$ flows through the residual softmax (below). The `topk` subgradient in (3) sends the dual pressure to exactly the worst-$(1-\alpha)$ weeks, and only while the constraint is breached. The penalty therefore pushes up the worst weeks and leaves the rest of the path largely alone, so the constraint binds once the dual is correctly scaled (Section 5).

Residual policy over an adaptive anchor. To generalise across regimes rather than overfit one price path, the weights are $w_t = \mathrm{softmax}(\log a_t + f_\theta(s_t))$, where a_t is an adaptive inverse-volatility anchor computed from recent returns. At initialisation $f_\theta \approx 0$, so the policy starts as the anchor, a regime-robust prior, and training then learns a state-dependent tilt. Two choices curb backtest overfitting. Model selection keeps the checkpoint with the best validation objective among those that satisfy the validation CVaR limit, and weight decay is applied.

Baselines. The deterministic baselines are cash, equal weight, inverse volatility, minimum variance, mean–variance, risk parity and a min-CVaR Rockafellar–Uryasev LP [24], all re-estimated weekly. The covariance-based rules use the sample covariance of the trailing 104 weeks, and inverse volatility the trailing 52 weeks, and all are long-only under the learner's weight cap. Within each evaluation window, they are re-estimated from that window's returns, with a minimum of 26 weeks of history (equal weight before that). The learned baselines are model-free A2C, PPO [25] (clipped surrogate, generalised advantage estimation, GAE), SAC [14] (off-policy, twin critics, auto-tuned entropy), and the unconstrained differentiable allocator ($\lambda = 0$).

Regime-switching hybrid. We build a hybrid that needs no retraining. Each out-of-sample (OOS) week is labelled *high-volatility* if trailing 13-week SPY volatility meets the 75th-percentile cutoff (15% of OOS weeks), *selloff* if cumulative return is $\le -5\%$, and *calm* otherwise. The hybrid applies constrained RL in high-volatility weeks and rolling minimum variance otherwise, reusing the existing OOS paths without refitting. On the 26-week macro walk-forward, this labels 80 of 520 weeks as high-volatility, and in those weeks constrained RL has Sharpe 1.415 against 0.564 for minimum variance (Table 5). These labels differ from the regime split in Table 4.

Threshold sensitivity. The switch depends on two labelling thresholds, so we sweep them over vol-quantile $\in \{0.65, \ldots, 0.85\}$ and selloff $\in \{-0.04, -0.05, -0.06\}$, giving 15 configurations with no retraining (Table 1). The hybrid beats pure minimum variance in 10 of 15 configurations, with a mean gain of +0.077. It wins at every volatility cutoff when the selloff threshold is -0.04 or -0.05, but loses at all five cutoffs once the threshold deepens to -0.06. That threshold relabels as selloffs the high-volatility weeks in which the constrained allocator adds most, and routes them

Table 1: Sharpe gain of the hybrid over minimum variance across 15 threshold settings, with no retraining. The default is $(0.75, -0.05)$.

selloff \ vol. quantile	0.65	0.70	0.75	0.80	0.85
−0.04	+0.202	+0.230	+0.161	+0.225	+0.058
−0.05	+0.154	+0.179	+0.107	+0.169	+0.002
−0.06	−0.007	−0.017	−0.086	−0.034	−0.183

to minimum variance instead. The default $(0.75, -0.05)$, fixed before the sweep, gives +0.107, and seven of the ten winning configurations give a larger gain.

5 Decomposing the constraint into the dual, the architecture and budget feasibility

This section asks what the Lagrangian penalty contributes. It reports three findings, then the filter that addresses the third. We compare four versions, or *arms*, of the differentiable allocator (Table 2, top block). They are A1 unconstrained, A2 mis-scaled dual, A3 scaled dual, and A4, the scaled dual with a loose budget. All four share the residual actor, weight projection and budget (5 seeds, 1500 updates), and A2–A4 also share the CVaR penalty and CVaR-limited checkpoint selection. A2 and A3 differ only in the dual step, and A4 differs from A3 only in d. The mis-scaled dual updates λ with step $\eta_\lambda = 0.001$, far too small for the size of the CVaR excess, while the scaled dual uses $\eta_\lambda = 5.0$, which matches it. The unconstrained arm drops the penalty and selects checkpoints on validation return alone. The stress window trains on 2008–2018 and tests from late 2019 to 2024. Here the *rolling breach rate* is the share of test weeks in which the rolling 52-week $\mathrm{CVaR}_{0.95}$ exceeds the training budget d, so a rate of 1.000 means over budget in every test week.

Finding 1. A scale mismatch leaves the dual inert. With the mis-scaled dual (A2), λ stays near zero throughout training (Figure 2), with a mean final value of 0.022 against 0.101 for the scaled dual (A3). The dual update is $\lambda \leftarrow \max\{0,\ \lambda + \eta_\lambda(\widehat{\mathrm{CVaR}} - d)\}$, and the CVaR excess is about 10^{-2} in weekly-return units, so with a step of 10^{-3} each update moves λ by roughly 10^{-5}, and 1500 updates are not enough. The model-free actor–critic failed in the same way, through its *advantages* (how much better an action did than the critic expected). Its reward advantage was standardised to unit scale while its cost advantage stayed at about 10^{-3}, so the constrained and unconstrained agents trained identically. After we standardised the cost advantage and drove the dual with a breach indicator, λ rose to $O(1)$. As a check, if λ stays at zero throughout training, the constraint has no effect, whatever the loss curve looks like.

Finding 2. Most of the tail control comes from the pipeline. Most of the tail reduction does not depend on the penalty. Even with λ close to zero, the constrained pipeline lowers CVaR_{99} by 61% relative to the unconstrained learner (0.025 vs 0.065, 5-seed mean) and raises Sharpe by 58% (0.987 vs 0.626). The reduction comes from checkpoint selection under the CVaR limit, on top of the shared anchor and projection. A permutation test on the 5-seed distributions confirms the difference (observed −0.039, one-sided $p = 0.0044$, two-sided $p = 0.0083$, $n = 50{,}000$), and the scaled dual gives a similar reduction ($p = 0.0037$). Engaging the dual properly ($\lambda = O(0.1)$) barely changes CVaR_{99} (0.028 vs 0.025), so most of the 61% comes from the pipeline. The pipeline thus controls the tail without help from the penalty, which also explains why the coupling failure goes unnoticed. Tail metrics improve whether or not the dual works, and only the trace of the multiplier shows that it is inactive.

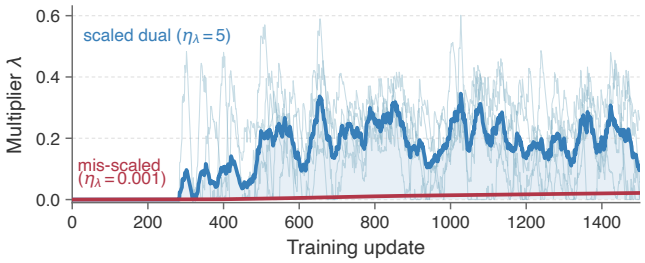


Figure 2: The Lagrange multiplier over training. Thick lines are five-seed means and thin lines the individual seeds. λ is zero while the anchor-initialised policy meets d.

Finding 3. Meeting the budget out of sample is a separate problem. Scaling the dual fixes training but not the test window. The scaled dual binds in training, but at the tight budget ($d = 1.2\%$) it breaches more often out of sample than the mis-scaled dual (75% of test weeks against 8%), although classical rules meet the same budget in 92% of weeks (Section 5). It is worse than the mis-scaled dual on every metric in Table 2.

Checkpoint selection explains why. With the dual scaled, most checkpoints meet the validation budget (27 of 30 on average, against 8 of 30), so selection keeps a later checkpoint that leans further toward return (update 1010 on average, against 370), and that checkpoint breaches when the tail shifts in the test window. Changing d does not fix this. Four rules for setting d from validation data, fixed in advance, all give 71–78% out-of-sample breach. They are a loose 1.92% budget (A4), the policy's own or the anchor's validation $\mathrm{CVaR}_{0.95}$, and a per-fold adaptive budget over a 10-fold walk-forward. Whatever the value of d, a Lagrangian learner whose dual binds gives up compliance when the tail shifts, so the remedy has to act on the portfolio itself.

The fix is a decision-time filter. We therefore enforce the budget at the moment of trading with a CVaR *filter*, a safety layer in the sense of Dalal et al. [10] that sits between the actor and the environment. For each proposed portfolio, it estimates forward CVaR from the trailing mean μ_p and volatility σ_p as $\mathrm{CVaR} \approx c_\alpha \sigma_p - \mu_p$, where $c_\alpha = \varphi(\Phi^{-1}(\alpha))/(1-\alpha)$ is the Gaussian CVaR multiplier ($c_{0.95} = 2.06$). If the estimate exceeds d, the filter moves the proposal along the line toward a trailing minimum-variance portfolio, using the largest mixing weight that clears the budget. The turnover, max-weight and long-only limits then apply as before. We train with the filter on.

The filter uses no future data. The trailing mean, covariance and minimum-variance anchor come from the $L = 104$ returns $\rho_{t-L}, \ldots, \rho_{t-1}$, and the actor's features use only $\rho_{<t}$, so the decision at t depends only on $\mathcal{F}_{t-1}$. An automated look-ahead test confirms that shifting the return series leaves the decision at t unchanged, with zero deviation.

Table 2: Coupling ablation and decision-time filter on the stress window (5-seed means). †Mean online-updated λ at evaluation.

variant	Sharpe	$CVaR_{99}$	breach rate	final λ
A1 unconstrained	0.626	0.065	1.000	0.000
A2 mis-scaled dual (η_λ=0.001)	0.987	0.025	0.079	0.022
A3 scaled dual (η_λ=5.0, d=1.2%)	0.911	0.028	0.751	0.101
A4 scaled dual, loose (d=1.92%)	0.543	0.050	0.780	0.046
A5 filter only ($\eta_\lambda = 0.001$)	0.987	0.025	0.079	0.003
A6 filter + scaled dual ($\eta_\lambda = 5.0$)	0.851	0.016	0.074	0.135
A7 adaptive dual ($\eta_\lambda = 5.0$ online)	0.826	0.031	0.819	7.692†
A8 filter + adaptive dual	0.800	0.020	0.146	0.657†

Four further arms, A5–A8, match the coupling arms exactly (5 seeds, 1500 updates, same splits) and change only the named quantities (Table 2, bottom block, and Figure 3). A6, the method we propose, adds the filter to the scaled dual. At the same η_λ, it lowers breach from 75.1% to 7.4%, below every classical rule, with $CVaR_{99}$ 0.016, mean excess 0.0043 and Sharpe 0.851. A5 adds the filter to the mis-scaled dual ($\eta_\lambda = 0.001$) and matches the mis-scaled row. That policy already sits within budget, so the filter never triggers (0 of 1285 test weeks across seeds). A5 reaches the classical 7.9% at Sharpe 0.987, against 0.901 for minimum variance, with breaches less than half as deep (mean excess 0.0126 vs 0.0294). A7 is the negative control, which updates λ online from realised losses with the filter off. Its breach rate stays at 81.9% even as λ climbs to a mean of 7.69, so adjusting the dual at test time does not enforce the budget, and the improvement in A6 is due to the filter. A8 pairs the filter with the adaptive dual and recovers most of the compliance (14.6%), though less reliably, ranging from 7 to 33% across seeds, whereas A5 and A6 do not vary between seeds.

The filtered arms hold low-volatility portfolios that differ measurably from minimum variance. A5's weights are a mean L_1 distance of 0.476 from rolling minimum variance, with correlation 0.786 and a different top holding in 9.4% of rebalances, against 0.890, 0.201 and 97.5% for the unfiltered scaled dual. In that region, A5 beats minimum variance on both axes, with higher Sharpe at an equal breach rate and half the breach depth, and A6 has a lower breach rate. Out of sample, the scaled soft penalty did not keep the portfolio within budget, and the decision-time projection (A6) did.

Is the budget infeasible, or is the policy failing to reach it? The budget can be met. When every classical baseline is re-scored against the identical rolling-$CVaR_{0.95}$ budget on the same realised path, the rolling breach rate is 7.9% for minimum variance, inverse volatility and the min-CVaR LP, 62.8% for equal weight and 75.3% for risk parity, against 75.1% for the scaled-dual RL. The soft-penalty learner did not meet it, and the filtered arms reach the classical rate with a learned policy.

6 Data

The primary universe consists of seven liquid exchange-traded funds (ETFs) that span the major macro blocks, namely SPY (equity), TLT (rates), HYG (credit), DBC (commodity), GLD (gold), UUP (US dollar) and BIL (cash). Adjusted closes from Yahoo are resampled to 887 weekly returns over 2008–2024. The splits are chronological (60/20/20), with train ≈2008–2018 (including the 2008 global financial crisis, GFC), validation ≈2018–2021 and test

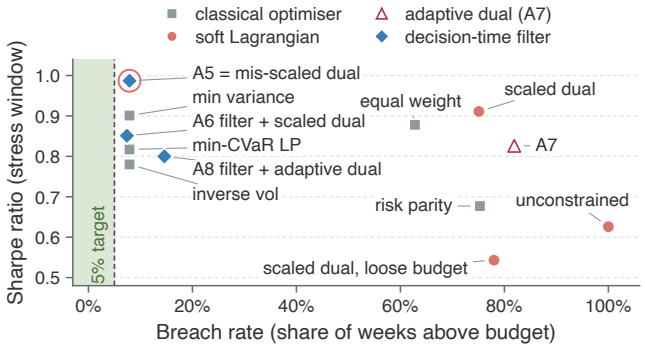


Figure 3: Breach rate against Sharpe on the stress window. The dashed line marks the 5% tolerated breach rate. Squares are classical optimisers, diamonds the filtered arms (A5, A6, A8), circles A1–A4 and the triangle the adaptive dual without the filter (A7). The ringed point is the mis-scaled dual, which A5 overlaps.

≈2021–2024. For the stress study, we retrain on 2008–2018 and test on the 267-week window containing the 2020 COVID crash and the 2022 selloff.

Macro covariates (10y−3m term spread, VIX) are sourced from index proxies (`^TNX`−`^IRX`, `^VIX`), and a Federal Reserve Economic Data (FRED) loader is also provided. Gross exposure is 1.0, with the weight and turnover caps of Section 3. Training uses 150 episodes and seeds {7, 13, 23, 42, 2025}. Significance uses a 2000-sample block bootstrap (block 4), Wilcoxon, Benjamini–Hochberg and Deflated Sharpe.

For the cross-universe significance test (Section 7.4) we add a second universe of ten S&P sector ETFs (XLK, XLF, XLE, XLV, XLY, XLP, XLU, XLI, XLB, plus a BIL cash leg), identical in every other respect (same provider, clock, window and cost). The two universes share a calendar, and Section 7.4 clusters their folds accordingly. For the evaluation-protocol experiment (Section 7.8) we add a third universe of 31 cross-asset, sector, and industry ETFs (4 broad equity, 9 S&P sectors, 7 industry, 6 fixed-income, 3 commodity, FX, cash) with the same provenance, clock and cost.

7 Results

Sharpe ratios for the constrained allocator (0.85 to 0.99) differ across tables because configurations differ. The coupling and filter arms use 1,500 updates over 5 seeds, the stress study 2,500 over 5 seeds, the ablations 2,000 over 3 seeds, and each walk-forward fold a 3-seed ensemble.

7.1 Calm chronological split, where the constraint does not bind

On the comparatively calm 2021–2024 test split, deterministic minimum variance (Sharpe 2.52, $CVaR_{95}$ 0.0064) and the min-CVaR LP (2.24) dominate. The differentiable allocator reaches Sharpe 1.24 (vs equal weight 0.82, inverse vol 1.45), and model-free A2C reaches only ≈0.79. No strategy breaches the CVaR limit, so the constrained and unconstrained learners are nearly identical, and the constraint costs almost nothing when there is no tail loss to prevent. The other deterministic rules trail on this split, with

Table 3: Unconstrained and constrained allocators on the stress window (mean ± std over 5 seeds).

metric	unconstr.	constr.	change
Sharpe	0.63±0.09	0.88±0.04	+40%
Sortino	0.61	0.87	+44%
Calmar	0.32	0.48	+50%
Max drawdown	0.206	0.118	−43%
$CVaR_{95}$	0.0333	0.0189	−43%
$CVaR_{99}$	0.0647	0.0327	−49%
Limit breach rate	0.234	0.189	−19%
Violations	7.2	0.0	−100%

Sharpe, $CVaR_{95}$ and max-DD of 1.45, 0.0087 and 0.041 for inverse volatility, 0.84, 0.0198 and 0.104 for mean–variance, 0.82, 0.0163 and 0.097 for equal weight, and 0.70, 0.0182 and 0.134 for risk parity. Minimum variance and the min-CVaR LP share a max-DD of 0.016, and the LP has $CVaR_{95}$ 0.0080.

7.2 Stress window, where the constraint binds

Table 3 contrasts a return-greedy unconstrained learner with the CVaR-constrained allocator on the stress window (mean ± std over 5 seeds). The constraint roughly halves tail risk and drawdown and improves Sharpe, Sortino and Calmar, because it removes the deep-loss weeks. Under stress, the constrained policy moves into the cash and rates assets, consistent with its lower drawdown and breach depth. In this table, the *limit breach rate* uses the environment's 3% limit as d (the rolling breach rate of Table 2 uses the 1.2% training budget), and violations count the weeks per run in which a portfolio constraint is breached.

A paired block bootstrap of the $CVaR_{99}$ difference (constrained minus unconstrained, 2000 resamples, block 4) gives −0.033, with 95% bias-corrected and accelerated (BCa) interval $[-0.035, -0.022]$ and $p \approx 0$, so the tail-risk reduction is statistically significant. We report BCa rather than percentile intervals because resamples of an extreme-tail statistic often omit the few crash weeks, which biases a percentile interval toward zero. The constrained allocator also eliminates portfolio-constraint violations (7.2 → 0.0 per run).

7.3 Walk-forward across regimes (10 OOS folds, ≈2010–2024)

We refit every 52 weeks on a rolling 312-week window and concatenate 10 out-of-sample folds. Out of sample, the constraint beats unconstrained RL on every metric, with Sharpe +18% (0.70 → 0.83), $CVaR_{99}$ −20% (0.0416 → 0.0331) and max-DD −25% (15.1% → 11.4%). In absolute terms, inverse volatility (Sharpe 1.06, max-DD 0.070, $CVaR_{99}$ 0.0183) and minimum variance (1.45, 0.057, 0.0106) remain ahead, and minimum variance is the strongest of all, so on this protocol the learner does not beat the optimiser. On the macro universe alone, the fold-level Wilcoxon on $CVaR_{99}$ is marginal ($p = 0.06$, with 5 of 10 folds improved), and the reduction concentrates in selloffs ($CVaR_{99}$ 0.066 → 0.048, max-DD 0.218 → 0.137, Table 4). A two-universe, 40-fold analysis raises the significance of the tail reduction to $p = 8.8 \times 10^{-6}$ (Section 7.4).

When RL adds value, by regime. The constrained policy does best relative to minimum variance in high-volatility periods (n=71), with Sharpe 2.169 versus 1.668 and $CVaR_{99}$ within 0.012 of minimum variance's floor. In calm markets (n=388) and selloffs (n=61), minimum variance dominates (selloff Sharpe 0.035 vs. constrained RL −0.809). The overlay adds value when volatility is elevated but has not collapsed into a full drawdown, and stops adding value once liquidity deteriorates. The constraint still cuts tail risk relative to unconstrained RL in all three regimes (−29.1% calm, −20.7% high-vol, −27.4% selloff), which suggests that the reduction comes from the pipeline and does not depend on the regime.

Table 4: Sharpe and $CVaR_{99}$ by regime, pooled from the 10-fold macro-ETF walk-forward (n is the number of weeks in each regime).

	Calm (n=388)		High-vol (n=71)		Selloff (n=61)	
Strategy	Sharpe	$CVaR_{99}$	Sharpe	$CVaR_{99}$	Sharpe	$CVaR_{99}$
RL Constrained	1.438	0.020	2.169	0.018	−0.809	0.048
RL Unconstrained	1.282	0.028	2.194	0.023	−1.127	0.066
Min-Variance	1.770	0.009	1.668	0.007	0.035	0.016
Inverse-Vol	1.721	0.011	2.803	0.009	−1.327	0.033

Table 5: Regime-switching hybrid against the pure strategies on the 26-week macro walk-forward, with bootstrap 95% intervals from 50,000 resamples.

Strategy	Sharpe	95% CI	Ann. Ret.	$CVaR_{99}$
Constrained RL (pure)	0.942	[0.29, 1.68]	4.0%	0.025
Min-variance (pure)	1.209	[0.55, 1.93]	2.9%	0.012
Hybrid	1.316	[0.66, 2.03]	3.5%	0.013

7.4 Statistical significance across two universes

To check that the effect is not specific to one universe, we add the ten-sector universe (Section 6) and shorten the walk-forward test fold to 26 weeks, which gives 20 folds per universe (Table 6). Each universe shows a significant fold-level reduction on its own, with macro-ETF Wilcoxon $p = 0.0077$ (9 folds improved, 11 tied, none worse) and sector $p = 0.0003$ (17 improved, 2 tied, 1 worse), and Benjamini–Hochberg [6] multiple-testing control rejects the null for both. A tie is a fold where the budget never binds, so the two learners coincide. The mean fold $\Delta CVaR_{99}$ is −0.0041 (macro-ETF) and −0.0087 (sector). Pooling all 40 folds gives Wilcoxon $p = 8.8 \times 10^{-6}$ and a bootstrap 95 % CI of $[-0.0089, -0.0041]$ on the mean fold difference, and 26 of the 27 untied folds improve (sign test on directions only, $p = 4 \times 10^{-7}$).

The two universes share a calendar, so their folds are not independent. We also average their differences within each calendar fold. The 20 resulting clusters give Wilcoxon $p = 3.3 \times 10^{-4}$ (exact sign-flip test $p = 3.8 \times 10^{-5}$), and moving-block bootstrap intervals, which resample runs of consecutive clusters to keep time order, exclude zero. The Probabilistic Sharpe Ratio [4] (the probability that the true Sharpe is above zero) of the constrained allocator is ≥ 0.99 on both universes. Its out-of-sample Sharpe advantage over classical baselines, however, does not survive the Deflated Sharpe Ratio [5], which discounts a Sharpe ratio for the number of configurations tried before reporting it. The significant effect is on tail risk, which is what a safety layer is meant to deliver.

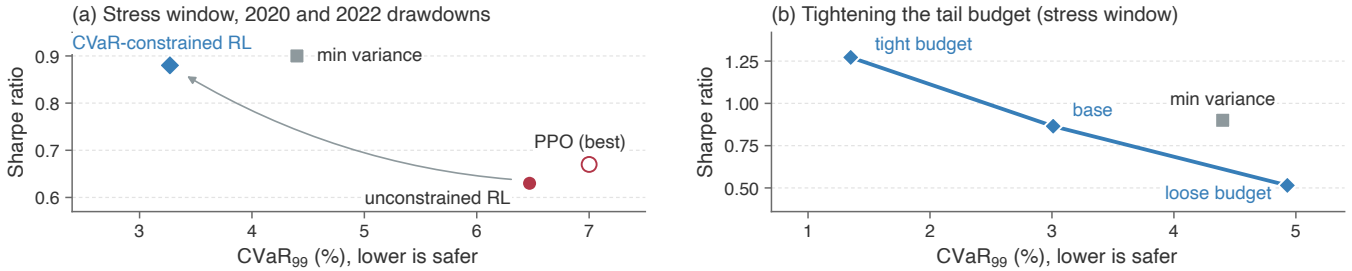


Figure 4: (Left) Risk and return on the stress window for the unconstrained (circle) and constrained (diamond) allocators. (Right) Tightening the tail budget from loose to tight improves both $CVaR_{99}$ and Sharpe. The square marks minimum variance.

Table 6: Walk-forward $CVaR_{99}$ on two universes with 26-week folds. PSR is the Probabilistic Sharpe Ratio of the constrained allocator.

universe	folds	$CVaR_{99}^{unc}$	$CVaR_{99}^{con}$	improved	Wilcoxon p	PSR
macro ETF (7)	20	0.0143	0.0102	9/20	0.0077	0.997
sector (10)	20	0.0336	0.0250	17/20	0.0003	0.990
pooled	40	—	—	26/40	8.8×10^{-6}	—

7.5 Generalisation to sector ETFs

Re-running the stress study unchanged on the ten-sector universe (mean over 3 seeds) replicates the main effect. The constraint cuts $CVaR_{99}$ by 30% (0.109 → 0.076), $CVaR_{95}$ by 35% (0.058 → 0.037), maximum drawdown by 30% (0.295 → 0.208), and the breach rate by 47% (0.75 → 0.40), at no Sharpe cost (0.752 → 0.748). A paired block bootstrap of the $CVaR_{99}$ difference gives −0.044, with 95% BCa interval $[-0.050, -0.032]$ and $p \approx 0$. The one difference from the macro universe is that there the constraint improved Sharpe (the avoided tail losses outweighed the cost), whereas on sectors it leaves Sharpe unchanged. The reduction in tail risk and drawdown carries over to sectors.

7.6 Robustness and ablations

The $CVaR_{99}$ reduction holds in all 7 perturbations, even at 3× costs ($CVaR_{99}$ 54% lower, Sharpe 63% above unconstrained). Ablations show a consistent pattern (Figure 4, right, with 3-seed means on the macro-ETF stress window). Tightening the tail budget from loose ($\alpha = 0.95$, limit 1.92%) through base (1.2%) to tight ($\alpha = 0.99$ or 0.72%) improves $CVaR_{99}$ and Sharpe together at every step. The tightest configuration reaches Sharpe 1.27 and $CVaR_{99} = 1.4\%$, above minimum variance on the same window (Sharpe 0.90, $CVaR_{99} = 4.4\%$) and competitive with its walk-forward Sharpe of 1.45. Here tighter limits also raise return, which risk limits usually do not, because the constraint acts only on the worst weeks, through its tail-only gradient and tighter checkpoint selection, and so removes the largest losses while leaving most of the return distribution unchanged. Against the base configuration (Sharpe 0.865, $CVaR_{99}$ 0.0301, max-DD 0.111), removing the anchor gives 0.826, 0.0318 and 0.108, and $\alpha = 0.90$ gives 0.684, 0.0452 and 0.172. A Sharpe objective gives 0.791, 0.0273 and 0.108, the loose budget 0.515, 0.0493 and 0.230, $\alpha = 0.99$ gives 1.251, 0.0146 and 0.053, and the tighter budget 1.272, 0.0135 and 0.041.

Table 7: Best-tuned model-free RL (selected on validation Sharpe, 3-seed mean) against the differentiable allocators and minimum variance on the stress window.

strategy	Sharpe (↑)	$CVaR_{99}$ (↓)	max-DD (↓)	turnover (↓)
minimum variance	0.90	0.044	0.113	0.011
diff. constrained	0.91	0.032	0.113	0.039
diff. unconstrained	0.60	0.062	0.212	0.023
PPO (best)	0.67	0.070	0.225	0.028
PPO (best) + CVaR	0.63	0.047	0.160	0.162
SAC (best)	0.39	0.033	0.185	0.005
SAC (best) + CVaR	0.18	0.035	0.205	0.004

Features. Rolling factor betas alone slightly hurt performance. Adding macro state gives Sharpe +0.02 (3-seed mean) but worsens $CVaR_{99}$ (0.029 → 0.031).

Does the policy change its holdings? Three diagnostics show that the constraint changes what the policy holds, so the lower tail metric is not a measurement artefact. The unconstrained allocator concentrates and holds through the 2020 and 2022 drawdowns. The constrained one rotates into cash and rates and keeps a lower, steadier rolling CVaR estimate. The tail control costs modestly higher turnover, with no leverage or single-asset bets, and the policy stays long-only, diversified and within the turnover cap.

7.7 Tuned model-free RL remains weak

We sweep 36 PPO and 8 SAC configurations, select each by validation Sharpe, and evaluate on the stress window (Table 7). Tuning does not help either family. The selected PPO overfits validation (Sharpe 1.70) and then has the worst out-of-sample tail (Sharpe 0.67, $CVaR_{99} = 0.070$), while SAC collapses to near-zero turnover (Sharpe 0.39). With 887 weekly observations, score-function and off-policy RL have too few effective samples. The constrained allocator matches minimum variance on Sharpe (0.91 vs 0.90) and lowers $CVaR_{99}$ by 27% (0.032 vs 0.044).

7.8 Evaluation-protocol reversal

To test how much the choice of evaluation protocol alone can change the conclusions, we use the 31-asset universe (Section 6). Covariance estimation is harder here, so a learner has, in principle, more room to add value. On a single stress split (train pre-2018, test through 2024), the constrained learner appears to beat every optimiser, with Sharpe 0.85 against 0.23 for minimum variance (Table 8). Under a rolling walk-forward (20 out-of-sample folds, refit every 26 weeks), the same minimum-variance optimiser

Table 8: Sharpe ratios by protocol, same 31-asset universe.

protocol	min-var	inv-vol	constr. (ours)	unconstr.
stress split	0.23	0.38	0.85	0.70
walk-forward (OOS)	1.40	0.78	0.74	0.56

is instead the best method by a wide margin, with Sharpe 1.40 against the learner's 0.74, so the ranking inverts.

The stress-split result depends on one fixed period and disappears under rolling refits. Covariance shrinkage does not explain it. On the same split, Ledoit–Wolf minimum variance [19] reaches Sharpe 0.25 and the min-CVaR LP 0.26. The tail-risk effect holds under both protocols (the constraint still lowers CVaR_{99} against the unconstrained learner fold by fold, Wilcoxon $p = 6 \times 10^{-4}$), while the apparent advantage over the optimiser does not. We recommend rolling walk-forward with Deflated-Sharpe [5] as the reporting standard, and treat single stress-window comparisons of learned allocators against optimisers as unreliable.

8 Discussion

How close is the learner to minimum variance? It depends on the arm. The unfiltered learner is far from it. Its mean per-rebalance L_1 weight distance to the rolling min-variance path is 0.89 (of a maximum 2.0), with correlation 0.20 and a different top-weighted asset in 97% of weeks. The similarity suggested by the tightest ablation (Figure 4, right) is in aggregate risk statistics and does not extend to holdings. The filtered arms are closer (L_1 0.48 to 0.61, correlation 0.72 to 0.79). They achieve compliance partly by holding low-volatility portfolios, and from there they improve on minimum variance in breach depth, and A5 also in Sharpe (Section 5).

We draw three practical lessons. First, tail limits reduce risk before a crash, as the constrained allocator moved to cash two weeks before the COVID trough (drawdown 7.7% against 16.3% unconstrained in March 2020). Second, the constraint is most useful as an overlay in high-volatility periods (Table 5). Third, a multiplier stuck near zero means the constraint is inactive, so check the coupling and use out-of-sample folds before claiming an edge over an optimiser.

9 Scope

The pure learned allocator does not beat rolling minimum variance on Sharpe out of sample, a Deflated-Sharpe correction removes its apparent Sharpe edge over classical baselines, and the hybrid's +0.107 gain comes only from high-volatility weeks. The study covers two equity-centric universes at weekly frequency with proportional costs (no market impact or borrow), the macro features are lagged for availability but are not vintage data, and the protocol reversal is shown on a single 31-asset window. The filter's Gaussian CVaR estimate can understate fat-tailed losses, and the constrained learner trails minimum variance in selloff weeks (Table 4). Within these limits, the paper contributes a tail-risk safety layer for learned allocators and evaluation lessons that transfer to other constrained RL studies in finance.

10 Conclusion and future work

Our differentiable CVaR-constrained allocator consistently reduces tail risk (26 of 27 untied folds, $p = 8.8 \times 10^{-6}$), removes portfolio-constraint violations under stress and reduces risk before crisis troughs, and a regime-switching hybrid reaches Sharpe 1.316 against 1.209 for min-variance (with overlapping intervals). We describe two failure modes that apply to constrained RL in finance generally, a Lagrangian constraint-coupling failure and an evaluation-protocol reversal, and show that a decision-time CVaR filter achieves the out-of-sample compliance that neither rescaling the dual nor recalibrating the budget achieved. Constrained RL is best used as a safety layer and an overlay for high-volatility periods. Its lasting benefits are lower tail risk and compliance with the budget.

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